MONASH University

# FIT3139 Final Project

**Ari Gestetner - 33966753**
ages0001@student.monash.edu
Bachelor of Commerce and Science.
FIT3139 - Computational Modeling and Simulation.

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TECHNOLOGY

**Faculty of Information Technology**

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# 1 Section 1: Specification table

| | |
|---|---|
| **Base Model** | The base model is the SIR model which describes the spread of a disease with births and deaths (workshop May 1) |
| **Extension Assumptions** | We will consider SIR with births and deaths with the addition of prediction intervals as well as the ability to set lockdown periods. |
| **Techniques Showcased** | Gillespie's Algorit, Monte Carlo, and heuristics. |
| **Modelling question 1** | Can we produce accurate prediction errors using the Gillespie algorithm |
| **Modelling question 2** | How can we plan a pandemic that we minimise lockdowns whilst also reducing infections and deaths |

# 2 Section 2: Introduction

Many real world events can be modelled using systems of ODE's, however, in the real world, events are rarely deterministic and often involve some form of randomness. Thus a better approach is to use stochastic models which can better align with the real world.

One issue with stochastic models is that we must run a simulation to analyse a single realisation of the model. Thus in order to look at the different possible outcomes assuming the model is correct, we must run multiple simulations of the model and look at the possible outcomes. Whilst this is useful for learning about outlier events (such as a possible event of extinction), there are times where we may want to learn about a range of possible outcomes.

In time series forecasting, we can fit a model to data in order to forecast future values based on the past. However, these values are just the mean value assuming the model fits the data correctly, they don't reflect the range of possible values which can be realised. A solution to this problem is to generate prediction intervals which is an "estimate of an interval in which a future observation will fall, with a certain probability, given what has already been observed" (Wikipedia 2025). These intervals are usually generated assuming normality which is not always possible. In cases where the models residuals are not normal, we can bootstrap various outcomes using simulation and use these values to generate the intervals (see *Forecasting: Principles and Practice*: Prediction intervals from bootstrapped residuals 2021).

With all this in mind, I wanted to know if we could achieve something similar with relation to dynamical models. This led me onto my first question;

> Can we use a stochastic dynamical model to construct prediction intervals for a deterministic dynamical model and if so, what can we learn from these intervals?

In order to answer that question, I needed to first choose a model, for which I chose the SIR model. After which I needed to construct the stochastic SIR model and run the Gillepsie algorithm various times and use the final outcomes to construct the intervals.

After selecting the SIR model, I then had another question,

> Can we use the evolutionary optimisation algorithm to find the optimal times to introduce lockdowns in order to minimise the time under lockdown whilst also attempting to maintain infections below some value?

This question had been answered for the base SIR model using constraint programming planning in “A Metric Hybrid Planning Approach to Solving Pandemic Planning Problems with Simple SIR Models” (Gestetner & Say 2024). However, over there, model solutions were required and lockdown changes (as in going from no lockdown to lockdown and vise versa) were limited to around 4 changes due to the difficulty of the problem. Nevertheless, by utilising heuristics, specifically the evolutionary optimisation algorithm, we can overcome these problems since we don’t require exact solutions as we can use Runge Kutta to estimate solutions to the model and we aren’t restricted to a small number of decisions (meaning we can implement lockdowns as many times as necessary).

By answering both these question we can gain insights into how diseases spread and how we can plan for them.

# 3 Section 3: Model Description

When looking at the model, we will first look at the base model, then we will consider both the extensions with relation to the questions they help answer.

## 3.1 Base SIR Model

The base SIR model considers the case where we have 3 compartments; Susceptible, Infected and Removed. These components are governed by the following system.

$$
\begin{aligned}
\frac{dS}{dt} &= -\gamma(S \cdot I) && (1)\\
\frac{dI}{dt} &= \gamma(S \cdot I) - \alpha \cdot I && (2)\\
\frac{dR}{dt} &= \alpha \cdot I && (3)
\end{aligned}
\tag{1}
$$

where

- $S(t) \in \mathbb{R}^+$ is the amount of the population which are susceptible to the infection.
- $I(t) \in \mathbb{R}^+$ is the amount of the population which are infected.
- $R(t) \in \mathbb{R}^+$ is the amount of the population which have recovered.
- $\gamma \in [0,1]$ is the average number of susceptible individuals infected by one infectious individual per contact per unit of time (infection rate).
- $\alpha \in [0,1]$ is the average number of infected individuals recovered per unit of time (recovery rate).

## 3.2 SIR with Births and Deaths

One extension which we apply to the SIR model is adding births and deaths, this extension will be applied to both the stochastic SIR model and SIR with lockdowns. The dynamics are as follows;

$$
\begin{aligned}
\frac{dS}{dt} &= \beta(S + I + R) - \gamma(S \cdot I) - \delta(S)\\
\frac{dI}{dt} &= \gamma(S \cdot I) - \alpha(I) - \delta(I)\\
\frac{dR}{dt} &= \alpha(I) - \delta(R)
\end{aligned}
\tag{2}
$$

- $\beta$ is the birth rate and adds new individuals to the susceptible population at a rate proportional to the total population
- $\gamma$ is the rate at which susceptible individuals become infected due to contact with infected individuals
- $\alpha$ is the rate at which infected individuals recover and move to the recovered compartment
- $\delta$ is the death rate (assumed equal across compartments), removing individuals from all compartments proportionally

## 3.3 SIR Model with Prediction Intervals

### Stochastic SIR Model

Now using the simple extension we can model the deterministic SIR model as a stochastic model, which we can then use to create the prediction intervals.

To do this we need to define the following events,

$$\begin{aligned} e_1 =& \beta(S+I+R) && \text{Born into S} \\ e_2 =& \gamma(S \cdot I) && \text{get infected} \\ e_3 =& \delta(S) && \text{Die in S} \\ e_4 =& \alpha(I) && \text{Recover} \\ e_5 =& \delta(I) && \text{Die in I} \\ e_6 =& \delta(R) && \text{Die in R} \end{aligned} \tag{3}$$

where the coefficients are the same as defined in Section 3.2.

After modelling the SIR model using a stochastic dynamical model, we then can simulate a realisation using a variant of the Gillespie algorithm.

**Algorithm 1** Stochastic Simulation using the Gillespie Algorithm

```
1: Initialise stochastic model with initial states and event functions
2: Initialise current time as start time, t_curr = t_0
3: while current time is less than total simulation time do
4:     Compute the propensity p_i of each possible event e_i
5:     Compute total rate t_p = Σ p_i
6:     Sample X ~ Exp(t_p) as time until next event
7:     Increment t_curr = t_curr + X
8:     Randomly select an event to occur, weighted by propensity
9:     for each state variable in the system do
10:        Update the state according to the selected event
11:        Record the new value of the state at the current time
12:    end for
13: end while
14: return Simulation values
```

Algorithm 1

The algorithm works as follows. 1. We start by initialising time and and events. 2. We then use the events to compute the propensities at current time $t$ (using the simulated values at $t$). 3. After which we simulate the time of the next event as an exponential random variable using the sum of the propensities as the rate parameter. 4. Finally we select the event by allocating probabilities $\frac{p_i}{t_p}$ for each event $e_i$ which is used to update the solution.

In order to explain this algorithm, we must refer to Poisson processes.
Consider for $i \in [1,n]$, $X_t \sim \text{Poisson}(\lambda t)$ and $Y_t^{(i)} \sim \text{Poisson}(\lambda_i t)$ where $\lambda = \sum \lambda_i$. We can view $X_t$ as a merged process consisting of $\sum Y_t$. then from field of random processes (*Combining and Splitting Poisson Processes* 2022), we know that each arrival time of $X_t$ is of type $i$ with probability $p_i = \frac{\lambda_i}{\lambda}$.
Since Poisson processes are just a counting process with exponential inter-arrival times, we can use the same concept for an exponential random variable. Thus, we only need to simulate a single exponential random variable to know event should occur next.

### SIR Model with Prediction Intervals

Now that we have a stochastic model and the Gillespie algorithm which allows us to run a simulation of the model, we can use Monte-Carlo to run multiple simulations of the model and use the results to construct $\alpha$% prediction intervals.

The process of constructing these intervals uses the following algorithm;

**Algorithm 2** Interval Construction

1: **Input** Simulation results, state
2: Bin simulation results into uniform time intervals
3: **for all** $t \in$ TimeIntervals **do**
4: Get Lower and Upper quantile for $t$
5: **end for**
6: **return** Lower and Upper quantiles

Now, with these intervals we can use Runge Kutta to find a solution to the deterministic dynamical model and we can plot the solution on top of the prediction intervals and analyse the variation as well as convergence/divergence of the model.

## 3.4 SIR With Lockdowns

Alternative to analysing how the model behaves, we can also utilise heuristics in order to reason around disease spread and learn when are the optimal times to introduce or remove lockdowns.

The new extension to the SIR model is

$$\begin{aligned}
\frac{dS}{dt} &= \beta(S+I+R) - \gamma(t)(S\cdot I) - \delta(S) \\
\frac{dI}{dt} &= \gamma(t)(S\cdot I) - \alpha(I) - \delta(I) \\
\frac{dR}{dt} &= \alpha(I) - \delta(R) \\
\gamma(t) &= \begin{cases} \gamma_0 & \text{Not in lockdown period} \\ \gamma_1 & \end{cases}
\end{aligned}$$

Where $\gamma_0$ is the infection rate during lockdown and $\gamma_1$ is the infection rate with no lockdown.

In order to find an optimal lockdown plan, we must define a loss function which we would like to minimise, then we will describe the evolutionary algorithm used to find an optimal solution.

First, Let $\mathbb{1}_L(t)$ be the indicator variable for whether lockdown is being implemented or not. Additionally, let $\mathbb{1}_I(t,a)$ be the indicator variable whether infection are above some cap $a$, thus we have

$$\mathbb{1}_L(t) = \begin{cases} 1 & \text{During lockdown} \\ 0 & \text{Otherwise} \end{cases} \quad \mathbb{1}_I(t,a) = \begin{cases} 1 & \text{Infections} > a \\ 0 & \text{Otherwise} \end{cases}$$

Now we will define the loss function as follows;

$$\mathscr{L}(\texttt{plan}, \texttt{time}) = \left( \sum_{t=0}^{\texttt{time}} \mathbb{1}_L(t) \right) + \lambda \left( \sum_{t=0}^{\texttt{time}} \mathbb{1}_I(t, a) \right) \tag{4}$$

Where $\lambda$ is the penalty for infections being above $a$.

Now we can refer to the evolutionary algorithm to optimise the solution;

**Algorithm 3** Evolutionary Optimisation Algorithm

```
Initialise a population of candidate solutions
Evaluate the fitness of each individual
for each generation do
    Select the best individual to keep
    while new population is not full do
        Select at random one of the following actions:
        - Mutate a parent
        - Crossover two parents
        - Copy a parent
        Add resulting child to the new population
    end while
    Re-evaluate fitness for individuals with missing scores
    Replace the old population with the new one
end for

return the best individual found
```

Thus, with this algorithm, we must define a crossover function as well as a mutate function. In this model, the solutions are a list of integers where 0 represents no lockdown and 1 represents a lockdown in place. Additionally, each index $t$ of this list corresponds to time $t \to t+1$ , thus it can be viewed as each day can be in lockdown or not.

Now the crossover function just slice the 2 parent solutions at some arbitrary index then concatenate them together.

**Algorithm 4** Crossover Function

```
Given parent candidates, lst1 and lst2
Randomly select a split point within the length of the lists
if random coin flip is heads then

    return lst1[:split_point] + lst2[split_point:]
else

    return lst2[:split_point] + lst1[split_point:]
end if
```

Then, our mutation function simulates 2 geometrically distributed random values with parameter $p = \frac{10}{\texttt{time}}$ which tends to simulate values closer to each other than just using a uniform distribution. However, the indices are then shuffled over by $U \sim \text{DiscUnif}[0, \texttt{time})$ which helps avoid clustering the changes around a single region.

**Algorithm 5** Mutation Function

Given list `lst`, create copy `new_lst`
Draw $G_1, G_2 \sim \text{Geom}(p)$ with $p = \frac{10}{\text{len}(\texttt{lst})}$
Draw $U \sim \text{DiscUnif}[0, \text{len}(\texttt{lst}) - 1]$
Compute indices: $\texttt{idx1} = (G_1 + U)\%\text{len}(\texttt{lst}), \texttt{idx2} = (G_2 + U)\%\text{len}(\texttt{lst})$
Flip a coin: $\texttt{implement\_lockdown} \in \{0, 1\}$
**if** `idx1` $<$ `idx2` **then**
    Set `new_lst[idx1 : idx2]` $\leftarrow$ `implement_lockdown`
**else**
    Set `new_lst[idx1 : end]` $\leftarrow$ `implement_lockdown`
    Set `new_lst[0 : idx2]` $\leftarrow$ `implement_lockdown`
**end if**

**return** `new_lst`

# 4 Section 4: Results

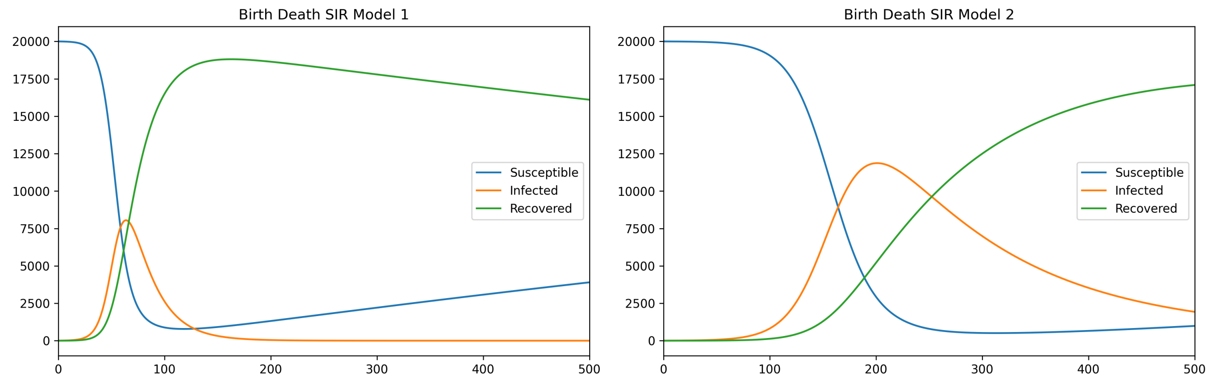
## 4.1 Birth Death SIR Solutions

We begin by first plotting the solutions to birth death SIR model using 2 sets of parameters in order to analyse their behaviours.

The below table lists both sets of parameters

| Parameters | Model 1 | Model 2 |
|---|---|---|
| $\beta$ | 0.0005 | 0.0005 |
| $\gamma$ | 0.00001 | 0.000003 |
| $\delta$ | 0.0005 | 0.0005 |
| $\alpha$ | 0.05 | 0.008 |

Additionally, our initial values are $S_0 = 20000$, $I_0 = 5$ and $R_0 = 0$.

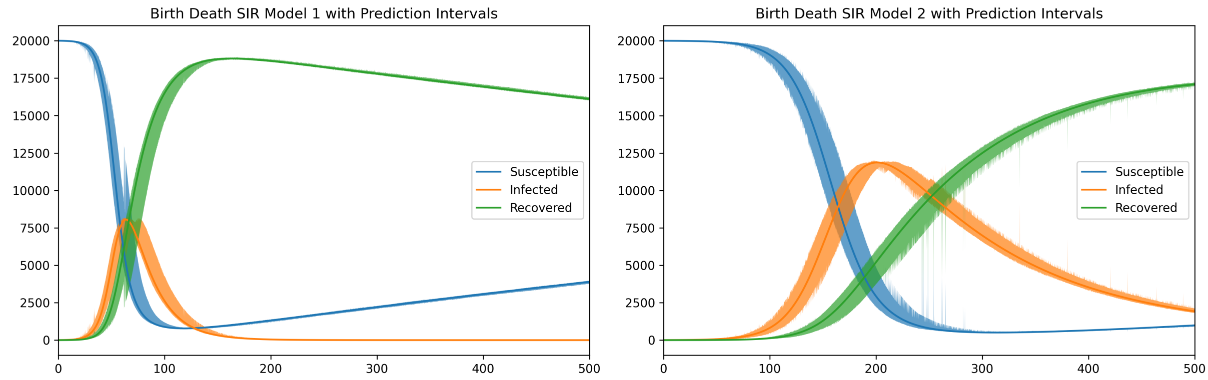



Now looking at the plots we can see that `model1` the infections peak a lot earlier on and extend for much longer than in `model2`. This is due to `model2` have lower infection rate $\gamma$ and recovery rate $\alpha$.

## 4.2 Prediction Intervals

Now with this, we can construct the prediction intervals with 80% quartiles using Monte-Carlo simulation and the Gillespie algorithm.

At first we can clearly notice that the interval bands on `model2` are clearly larger than in `model1`. This is likely caused by the longer period of high infections. When infections and recovered components




are small, then any events proportional to $I$ and $R$ would be less likely than events only proportional to $S$. Likewise, when $S$ and $I$ are small then the events only proportional to $R$ are most likely to occur. This would lead to small variance in these situations, which is why the bands are initially small as well as by the later periods once the model begins reaching a steady state.

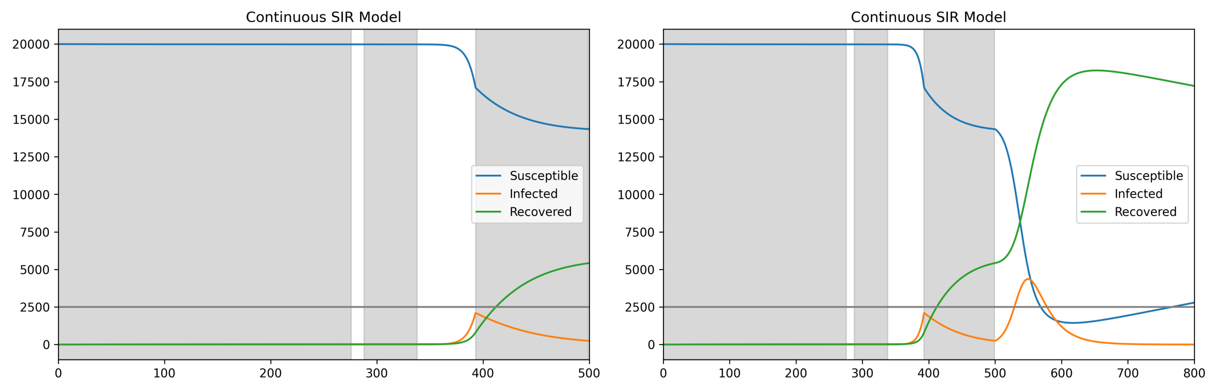
Furthermore, the primary reason for the large variation during the infection peak is due to all the components having similar values which leads to the events having much more evenly distributed probabilities of occurring, which tends to have much larger variation over each simulation. Thus, when the infection peak lasts longer, the variance increases.

Finally, we can explore the commonalities between the models. Like explained, the size of the bands initially begin quite small and grow as infections peak, however, once infections begin to reduce, so do the size of the bands. We can also note that the solutions to the SIR model tend to be skewed in the direction of its movement with respect to the bands. Meaning, that when a components values are decreasing, the solution tends to be closer to the lower quartile than the upper quartile. Similarly when the component is increasing, the solution tends to be closer to the upper quartile than the lower quartile.

## 4.3 Lockdown Planning

Now we will answer our second question by utilising the evolutionary algorithm to plan when lockdowns should be implemented.

The parameters of the model are $\gamma_0 = 1e{-}05$ when not in lockdown and $\gamma_1 = 2.0000000000000003e{-}06$, the infection cap $a = 2500$ with penalty $\lambda = 10$ as well as for the probabilities of mutation and crossover are $\mathbb{P}(\texttt{mutate}) = 0.25$ and $\mathbb{P}(\texttt{crossover}) = 0.25$, with a population size 10 over 30 generations.




**(a)** *Lockdown over planning horizon* **(b)** *Plan over extended horizon*

**Figure 1:** *Grey regions are lockdown periods and horizontal line is infections cap*

If we look at Figure 1a we can see that most time is spent under lockdown with the occasional period not having lockdown. We must note however, that since the planning domain was restricted to 500 we must refer to Figure 1b which extends time by 300.

Looking at his second plot, we can see that very soon after the planning domains time ends, there is a peak in infections which goes above the cap chosen.

This means, that this planning procedure may not be so useful if we want to plan over an undefined period of time. If we do know over how long we intend to plan for (say we only care about the next 2 years) then this may be a suitable algorithm.

## 5 Section 5: List of algorithms and concepts

- The SIR model with births and deaths as the basis for our models
- The use of Monte-Carlo simulation and the Gillespie algorithm to produce prediction intervals
- Runge Kutta to find the solution to the SIR equation
- An evolutionary algorithm was used to find an optimal lockdown plan

## 6 References

*Combining and Splitting Poisson Processes* (2022). [Online; accessed 2025-06-18]. `https://eng.libretexts.org/@go/page/44607`.

Gestetner, A & B Say (2024). *A Metric Hybrid Planning Approach to Solving Pandemic Planning Problems with Simple SIR Models*. arXiv: 2409.11631 [cs.AI]. `https://arxiv.org/abs/2409.11631`.

Hyndman, RJ & G Athanasopoulos (2021). *Forecasting: Principles and Practice*. 3rd ed. Accessed on 2025-06-17. Melbourne, Australia: OTexts. `https://otexts.com/fpp3/prediction-intervals.html#prediction-intervals-from-bootstrapped-residuals`.

Wikipedia (2025). *Prediction interval — Wikipedia, The Free Encyclopedia*. http://en.wikipedia.org/w/index.php?title=Prediction%20interval&oldid=1286934091. [Online; accessed 17-June-2025].

## 7 Generative AI Decleration

I have not used Generative AI for this assignment